Where does weight-space programming fix failures?

NVIDIA Cosmos3 Edge Policy (world-action model) and NVIDIA GR00T N1.7 DROID (VLA on a Cosmos-Reason2-2B backbone), with OpenPI π0.5 as the control, on the same RoboLab DROID tasks T1–T8 and a retention task R1.

Method. When a clearer wording of the same goal makes a robot model succeed, copy that wording's internal effect into the weights in closed form, one least-squares solve per layer: no training, and the instruction stays the same.

π0.5, the control (earlier records). Distillation fixed its two language failures: T1 27 → 106 of 128 unseen starts (81 fixed, 2 broke, $p = 7.2 \times 10^{-22}$) and T8 0 → 27 of 64; a random edit of the same size did nothing (9 vs 13 of 64); the T1 edit also lifted T2 and T3; no edit helped T4–T7.

NVIDIA models. Cosmos3 Edge Policy: T8 edit 0 → 56 of 64 unseen starts; GR00T N1.7: T1 edit 9 → 0 of 32 practice starts (the ones it was built from). First finding: GR00T N1.7 succeeded on 5 of 16 starts under RoboLab's performance render preset and 16 of 16 under the default preset, so all three are compared under the default. Same tasks, slots, metrics and statistics; this brief refreshes as runs finish.



Cosmos3 Edge Policy, T8: the wording gap; the clip plays on the live page.

What weight-space programming did on each task, on unseen starts unless marked practice

	Cosmos3 Edge Policy world-action model	**GR00T N1.7** reasoning-backbone VLA	**π0.5** OpenPI VLA, control
T1 Cube then banana	T8 edit: 10 → 12 of 16 practice starts 4 / 2; p = 0.69	9 → 0 of 32 practice starts 0 / 9, p = 0.0039	fixed: 27 → 106 of 128 earlier slots distilled edit; 81 / 2, $p = 7.2 \times 10^{-22}$; shared slots, unedited: 5 of 16
T2 Pick up the eyeglasses	T8 edit: 0 → 0 of 16 practice starts 0 / 0; p = 1	unedited 0 of 32 not edited	T1 edit: 22 → 44 of 64 earlier slots own edit 18 → 27 of 64, not significant
T3 Cube left of the bowl	T8 edit: 4 → 0 of 16 practice starts 0 / 4; p = 0.12	unedited 1 of 32 not edited	T1 edit: 8 → 36 of 64 earlier slots no own edit submitted
T4 Stack yellow on red	unedited 12 of 16 not edited yet	unedited 0 of 16 not edited	no gain earlier slots lookalike perception; no clear wording
T5 Mustard into the left bin	T8 edit: 8 → 0 of 16 practice starts 0 / 8; p = 0.0078	unedited 0 of 32 not edited	no gain earlier slots own edit 8 → 11 of 64; the grasp slips
T6 Stack three cubes	*running*	unedited 0 of 16 not edited	no gain earlier slots own edit 35 → 39 of 64
T7 Pick up the flat box	*running*	unedited 0 of 32 not edited	no gain earlier slots a missing skill; 0 of 64 unedited
T8 Mustard right of the raisin box, reference-first	0 → 56 of 64 56 / 0, $p = 2.8 \times 10^{-17}$	unedited 0 of 32 not edited	fixed: 0 → 27 of 64 earlier slots distilled edit; 27 / 0, $p = 1.5 \times 10^{-8}$
R1 Retention task	9 → 13 of 16 practice starts	unedited 28 of 32 not edited	kept: 32 → 32 of 32 earlier slots cube into the bowl, T1 edit
Rendering: performance → default preset	*running*	5 → 16 of 16 Banana onto plate	*running*
Random control, same size	control 0, unedited 0, edit 56 of 64 T8	control 11, unedited 9, edit 0 of 32 practice starts T1	9 of 64 earlier slots T1; unedited 13, edit 50
Where to edit	text pathway 56 of 64 T8, 64 unseen starts; unedited 0	Cosmos-Reason2 backbone 0, diffusion action head 0 of 32 T1, 32 practice starts; unedited 9	VLM 55, action expert 20, head 0 of 64 earlier slots T1; unedited 5; only the VLM edit transfers

	Cosmos3 Edge	**GR00T N1.7**	**π0.5**
T1 Cube then banana	10/16 · 63%	9/32 · 28%	5/16 · 31%
T2 Eyeglasses	0/16 · 0%	0/32 · 0%	18/64 · 28%
T3 Cube left of bowl	4/16 · 25%	1/32 · 3%	8/64 · 13%
T4 Stack yellow on red	12/16 · 75%	0/16 · 0%	13/32 · 41%
T5 Mustard into left bin	8/16 · 50%	0/32 · 0%	8/64 · 13%
T6 Stack three cubes	*running*	0/16 · 0%	35/64 · 55%
T7 Flat box	*running*	0/32 · 0%	0/32 · 0%
T8 Mustard, reference-first	0/16 · 0%	0/32 · 0%	0/64 · 0%
R1 Retention task	9/16 · 56%	28/32 · 88%	32/32 · 100%

Unedited success on each shared task; stronger shading is higher. Each cell gives successes of starts.

	Cosmos3 Edge	**GR00T N1.7**	**π0.5**
T1 Cube then banana	+2 / 16	**−9 / 32***	**+79 / 128***
T2 Eyeglasses	±0 / 16	not run	**+22 / 64***
T3 Cube left of bowl	−4 / 16	not run	**+28 / 64***
T4 Stack yellow on red	*running*	not run	+3 / 32
T5 Mustard into left bin	**−8 / 16***	not run	+5 / 32
T6 Stack three cubes	*running*	not run	+5 / 32
T7 Flat box	*running*	not run	±0 / 32
T8 Mustard, reference-first	**+56 / 64***	not run	not run
R1 Retention task	+4 / 16	not run	±0 / 32

gain, $p < 0.05$ · loss, $p < 0.05$ · no detectable change

* $p < 0.05$; outlined: the edit's own task

Each model's main edit, unchanged, on every suite task: edited minus unedited successes on the same starts. Teal: a gain; near-black (near-white in dark mode): a loss (exact McNemar $p < 0.05$); grey: no detectable change; the outlined cell is the task the edit was built for.

Live page with every table, figure and record: ali-adeeb.com/weightsmith/cosmos · Paper: ali-adeeb.com/weightsmith/paper.pdf · Code and records: github.com/adeeb10abbas/weightsmith

Draft for discussion. Simulation only (RoboLab, Isaac Sim), one released checkpoint per model, shared slots 30000–30031 and 96000–96063 with identical initial scenes; π0.5 cells are earlier records on their own slots until its shared-slot runs land. Exact McNemar tests on paired starts.